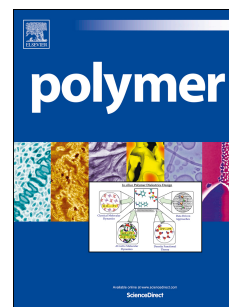


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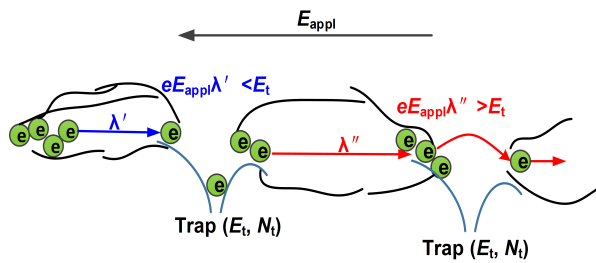
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Dielectric Relaxation and Trap-Modulated DC Breakdown of Polypropylene Blend Insulation

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Abstract

This work focuses on the dielectric relaxation of isotactic polypropylene (iPP), syndiotactic polypropylene (sPP) and atactic polypropylene (aPP) blended insulation. The relationship between dielectric relaxation and trap characteristics, as well as their effects on the breakdown properties of polypropylene insulation are investigated. Frequency domain spectroscopy (FDS) is employed to characterize the dielectric relaxation of PP blends at 50, 70 and 90 °C. Isothermal surface potential decay (ISPD) measurements are carried out to obtain the trap characteristics, and the dc breakdown strength is tested; thus, the mechanisms of trap characteristics on the dielectric relaxation and breakdown performance are analysed. The dielectric relaxation processes α and δ of PP blends can be attributed to segmental motions and carrier hopping, respectively. The α relaxation strength ($\Delta\epsilon_\alpha$) decreases with the addition of sPP, since the intermolecular interaction factor (g) between PP molecules is weakened. The activation energy of δ relaxation is proportional to the shallow trap level, since the increased shallow trap level can pose higher potential barriers for carrier hopping. The higher crystallinity of PP blends can induce more deep traps in the boundaries of spherulites, and it can reduce the number of shallow traps that are mainly distributed in amorphous regions. The increased deep trap density serves a function to reduce the mean free path of charges and restrain impact ionization, resulting in an enhancement in the breakdown field.

Keywords: polypropylene, blend insulation, dielectric relaxation, trap characteristics, crystallinity, dc breakdown

1. Introduction

Cross-linked polyethylene (XLPE) is widely used as the insulation material for high-voltage (HV) cables, owing to its superior insulation property, mechanical property and possibility to use at high temperature [1]. However, XLPE is not easy to be recycled at the end of its lifetime due to the cross-linked polymer network [2]. Polypropylene (PP) is considered to be a possible recyclable alternative to XLPE by virtue of its higher melting point of approximately 170 °C, high electrical resistivity and low dielectric loss [3-7].

Unfortunately, isotactic polypropylene (iPP) suffers from high brittleness and relatively low breakdown strength [8]. Blending iPP with syndiotactic polypropylene (sPP) has demonstrated a large enhancement in the breakdown performance and flexibility of iPP [5]. Atactic polypropylene (aPP) is a subsidiary product of the PP industry and has a poor electrical performance. However, iPP/aPP blends exhibit better ductility with aPP acting as a softening agent [9]. Exploiting PP blends with aPP at relatively low contents for high-voltage cable insulation is of great significance for both the economic and environmental factors. Accordingly, it is very necessary to study the electrical properties of

iPP/sPP/aPP blend materials and to estimate their practicability for HVDC cable insulation. Dielectric relaxation is usually considered to be one of the most important properties of a polymer [10]. The dielectric response at different frequencies and temperatures serves a function to characterize the molecule chain motion, dipole orientation and carrier transport properties of the polymer [19]. These characteristics provide a novel method to investigate a macroscopic electrical property from the microstructure view. Fuse *et al.* observed the four relaxation processes in polyamide nanocomposites by analysing their complex permittivity spectra: the dipole orientation, space charge polarization, interface polarization and group rotation, and it was found that the longer average relaxation time indicates the restriction of molecular motion, which should in turn suppress carrier transport [11]. Huang *et al.* investigated the molecular chain movement and carrier hopping relaxation of an epoxy resin and concluded that the enhanced size of free volume can provide enough space for chain movement and energy for carrier hopping [12].

Trap characteristics are effective when modulating the carrier transport and breakdown properties of dielectrics [13–15]. Deep traps might come from the boundaries of perfect, large spherulites, which provide the voids, cracks and low density regions that block the transport of charges [16–18]. Wang *et al.* found that deep traps induce more space charges or homocharges near the electrodes, the internal field generated by which results in less space charge accumulation in the bulk and reduced injection of hot-electrons [13]. It is also proposed that deep traps can capture carriers, resulting in a decrease of the carrier mobility and average free path of electrons. Thus, the breakdown strength of the polymer becomes enhanced [19, 20]. Kao *et al.* proposed that the trapping and recombination of charges in traps can create hot-electrons by the Auger process [21]. The breakdown starts due to the chain scission, which is caused by hot-electron impact.

In this study, iPP/sPP/aPP blends are prepared to investigate the effect of crystallization modulated trap characteristics on breakdown performance. A discussion based on dielectric relaxation is also provided to help in understanding the mechanisms of molecular motion and carrier hopping as they relate to trap energy.

2. Experimental setup

2.1 Sample preparation

Commercially available iPP (Oasis Petroleum Inc), sPP (Borealis) and aPP (Borealis) are used as the base material. The descriptions of the five PP blends, with different contents, studied in this investigation are shown in Table 1. The sample preparation procedure is as follows. First, the dried mixtures for the required polymer are put in a vacuum

oven at 80 °C for 48 h to ensure uniform initial thermal and mechanical conditions and eliminate cross linking by-products. Then the mixtures are placed into an internal mixer at a temperature of 205 °C and a speed of 30 r/min for 15 min in nitrogen atmosphere. At last, the obtained mixtures are hot pressed under 15 MPa at 205 °C for 30 min in a stainless-steel mould with dimensions of 90×90 mm². The thicknesses of the samples are 85 μm for the DC breakdown test and 200 μm for the frequency domain spectroscopy (FDS) and isothermal surface potential decay (ISPD) measurement.

Table 1. Index for PP blend samples investigated in this work.

Designation	iPP content	sPP content	aPP content
sPP0	95 wt%	0 wt%	5 wt%
sPP5	90 wt%	5 wt%	5 wt%
sPP15	80 wt%	15 wt%	5 wt%
sPP30	65 wt%	30 wt%	5 wt%
sPP45	50 wt%	45 wt%	5 wt%

2.2 Experimental methods

Differential scanning calorimetry (DSC) measurement of ~5mg samples is undertaken for five PP blends using a TAQ 2000 (TA Instruments, US). The detailed measurement procedure is described as follows. First, the samples are heated from 25 to 240 °C and held for 5 min to erase the thermal history. Then, the temperature is reduced from 240 to 25 °C to obtain the crystallization curve. Finally, the temperature is raised again from 25 to 240 °C to get the melting curve. The DSC measurement runs under nitrogen environment and at a temperature heating/cooling rate of 10 °C/min.

The ISPD method is employed to analyse the carrier transport and trap characteristics of PP blends. The schematic of the electrode arrangement and test circuit is presented in Figure 1. The corona discharge is conducted by a needle electrode connected to a HVDC source with a magnitude of ~8 kV. A grid electrode is arranged at 5 mm below the needle electrode and 5 mm over the surface of the sample.

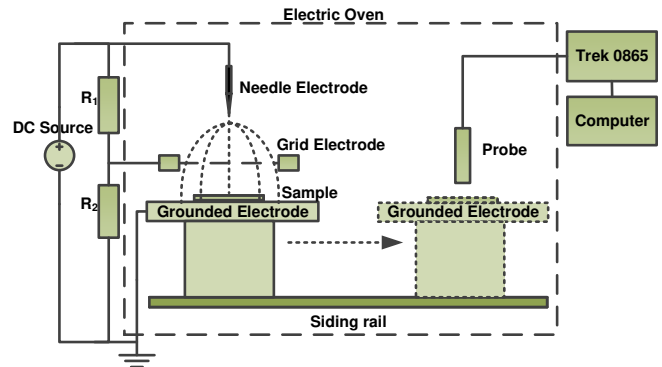


Figure 1. Schematic of electrode arrangement and test circuit of ISPD.

The ISPD test is performed in an electric oven where the relative humidity is held between 22%-30%. The charging time is set at 8 min. After the corona charging test, the sample is rapidly transferred to the underside of a Kelvin-type probe connected to an electrostatic voltmeter (Trek P0865) and the surface potential is recorded for 1800 s. The probe-to-surface distance is set at 3 mm.

In the FDS measurement, both sides of the samples are sputtered with aluminium electrodes, and the diameter of the electrodes is 90 mm. The complex permittivity is measured by a broadband dielectric spectrometer (Concept 40, Novocontrol) under 1 kV alternating voltage. The frequency in the test varies from 10^{-2} to 10^4 Hz and temperature is at 50, 70 and 90 °C.

The DC breakdown tests are performed in insulating oil at room temperature by using a set of ball-plate electrodes. The specimens, with a thickness of 85 μm and a diameter of 90 mm, are sandwiched between the electrodes and subjected to a rising voltage rate of 500 V/m until the specimens are broken down. Each specimen is tested 10 times, and the two-parameter Weibull distribution is employed to characterize the breakdown strength.

2.3 Trap level distribution based on the ISPD method

During the potential decay process, charge transport and dissipation are considered to be associated with the trapping and detrapping of carriers [22]. After the electric charges generated by the corona discharge transfer to the back electrode, as driven by the electrical field, the charges first become captured by carrier traps, then are detrapped by thermal activation and finally are transported to the grounded electrode, causing a decrease in the surface potential. The trap density can be calculated as follows [23, 24]

$$N_t(E) = \frac{4\epsilon_0\epsilon_r}{eL^2kT} \left| t \frac{dV_s(t)}{dt} \right| \quad (1)$$

where $N_t(E)$ is the trap density occupied by carriers at trap level E , e is electronic charge quantity, L is the thickness of sample in m, k is the Boltzmann constant, T is the absolute temperature, ϵ_0 is the permittivity of vacuum, and $\epsilon_r=2.18$ is the relative permittivity of PP.

The trap depth as a function of the decay time is defined as [23]

$$E_t = E_c - E_m = kT \ln(\nu t) \quad (2)$$

where ν is the escape frequency of trapped electrons, which is selected as 10^{12} s^{-1} in this test.

The ISPD curves are exponential functions of depolarization time. The surface potential can be well fitted by the bi-exponential function [24, 25]

$$V_s(t) = A_1 e^{-\frac{t}{\tau_1}} + A_2 e^{-\frac{t}{\tau_2}} \quad (3)$$

where V_s is the surface potential at time t , A and τ are the fitting parameters. The trap level distribution could be

decomposed into the shallow and deep traps, which represent the contributions of $A_1 \exp(-t/\tau_1)$ and $A_2 \exp(-t/\tau_2)$ respectively [24-26]. As a result, the shallow and deep trap levels and their densities can be obtained.

3. Results and analysis

3.1 Crystallinity

Figure 2 shows the DSC scans of five PP blends obtained during heating. The individual melting endotherms of sPP and iPP can be observed at ~ 130 °C and ~ 160 °C, respectively. However, there is no melting endotherm for aPP, since it is an amorphous polymer [27, 28].

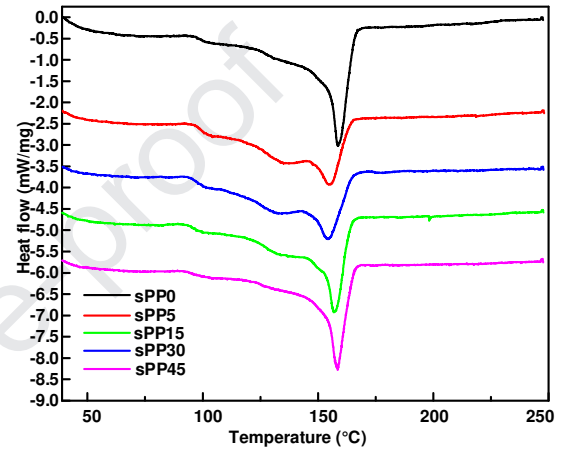


Figure 2. DSC curves of the heating scans for the five PP blends.

$X_{c,blends}$ is calculated from the DSC curves. The heat of fusion (ΔH_m^0) for 100% crystallinity of iPP is 209 J/g [29] and that of sPP is 196 J/g [30]. The ΔH_m^0 for 100% crystallinity of aPP is 0 J/g [27]. The formula to calculate the crystallinities of iPP/sPP/aPP blends is as follows [31]

$$\begin{aligned} X_{c,iPP} &= \Delta H_{m,iPP} / \Delta H_{m,iPP}^0 \times \text{wt\% iPP} \\ X_{c,sPP} &= \Delta H_{m,sPP} / \Delta H_{m,sPP}^0 \times \text{wt\% sPP} \\ X_{c,blends} &= X_{c,iPP} \times \text{wt\% iPP} + X_{c,sPP} \times \text{wt\% sPP} \end{aligned} \quad (4)$$

where wt%iPP and wt%sPP are the weight percents of iPP and sPP. There is not a term for aPP in the third of Equation 4 since it is non-crystalline polymer.

As a result, the crystallization temperature (T_c), melting enthalpy (ΔH_m) and crystallinity ($X_{c,blends}$) of the samples are shown in Table 2.

Table 2. Crystallization temperature, melting enthalpy and crystallinity level of PP blends measured by DSC tests.

Sample	T_c (°C)	ΔH_m (J/g)	$X_{c,blends}$ (%)
sPP0	119.44	78.82	39.69
sPP5	115.18	74.12	37.45
sPP15	115.90	75.89	38.60
sPP30	117.85	80.33	41.27
sPP45	119.55	79.08	41.04

3.2 Dielectric Constant and Dielectric Loss

Figures 3 and 4 show the real part (ϵ') and imaginary part (ϵ'') of the complex permittivity (ϵ^*) of the samples at frequencies ranging from 10^{-2} to 10^4 Hz and temperatures from 50 to 90 °C. In Figure 3, there are labelled two monotone rising trends of dielectric constant as the frequency decreases, representing the growing of the α and δ relaxation strengths, respectively. However in Figure 4, two relaxation peaks are observed in the relaxation spectra. First, the dielectric loss spectroscopy shows a relaxation peak above 10^2 Hz, and it can be identified as α relaxation process. The α relaxation is associated with the motion of segmental chains [12]. Moreover, as the frequency decreases, an increase in ϵ'' are

observed. These behaviours can be determined as δ relaxation processes that are involved with carrier hopping in localized states [12]. The peaks of α and δ relaxations both shift to higher frequency as the temperature rises, indicating that the more thermal energy is conducive to lower the average time of segmental chains' motion and carrier hopping in PP samples. In addition, when the temperature rises to 70 °C and 90 °C, the dielectric loss spectroscopy shows a new transition process below 10^{-1} Hz for 70 °C and 1 Hz for 90 °C. This new process also exists at 50 °C, but is too weak to be observed or totally covered by the adjacent δ relaxation peak. For this process, the peak intensity increases dramatically with temperature, and a

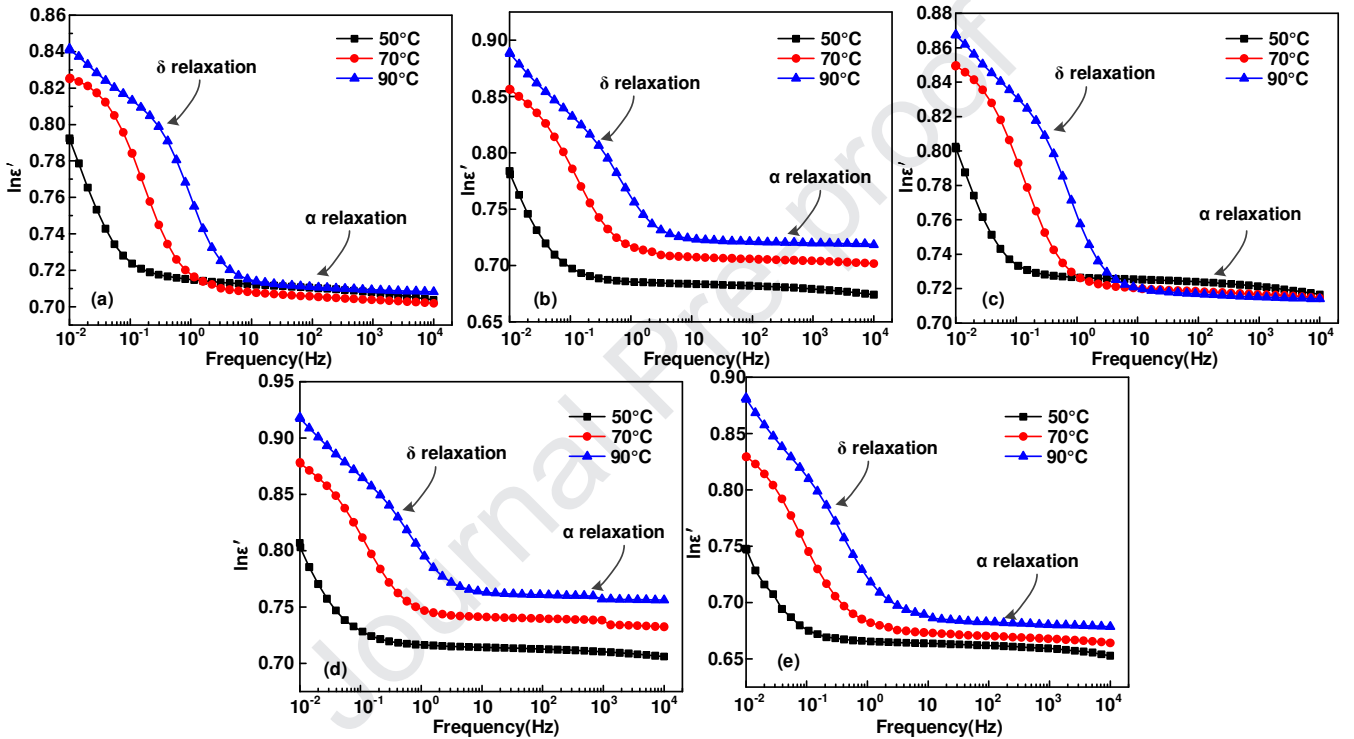


Figure 3. Real part of permittivity $\ln(\epsilon')$ as a function of frequency at 50, 70 and 90 °C for (a) sPP0, (b) sPP5, (c) sPP15, (d) sPP30, and (e) sPP45.

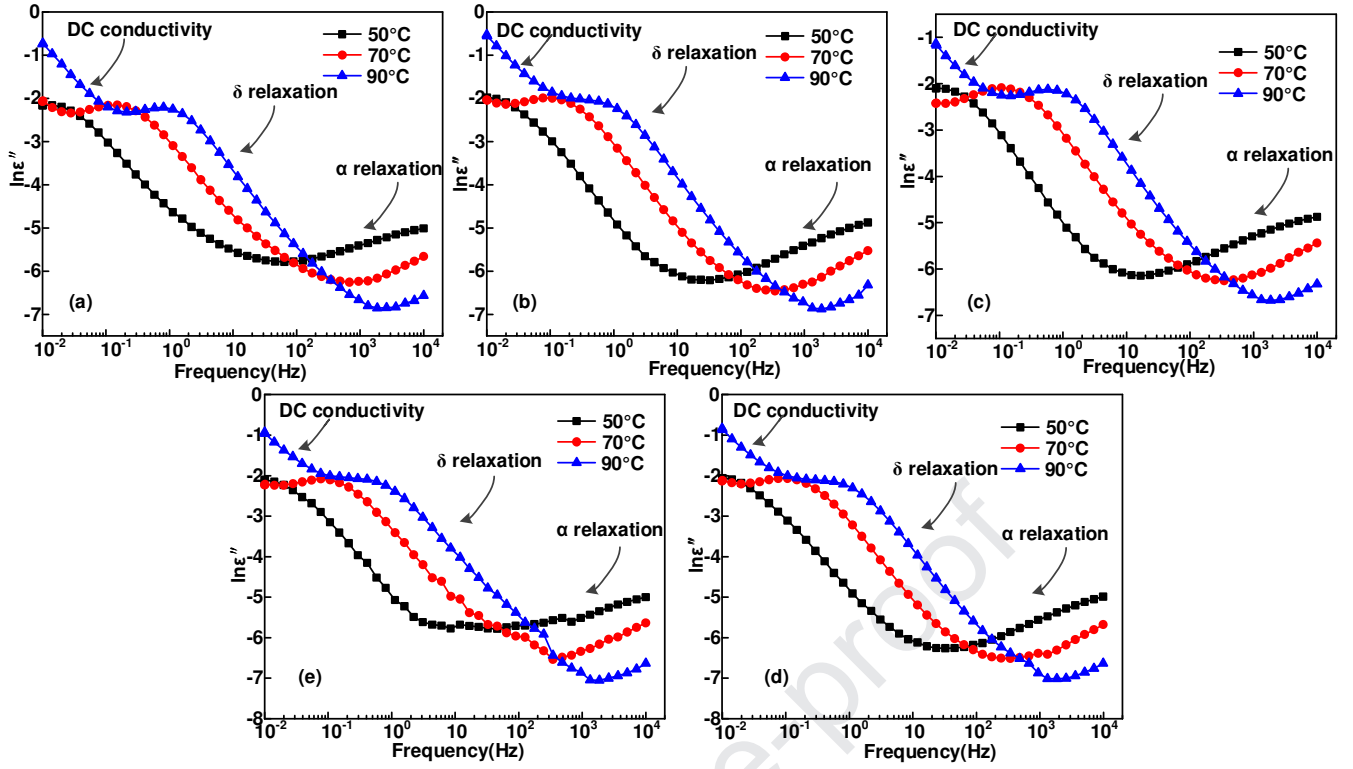


Figure 4. Imaginary part of permittivity $\ln(\epsilon'')$ as a function of frequency at 50, 70 and 90 °C for (a) sPP0, (b) sPP5, (c) sPP15, (d) sPP30, and (e) sPP45.

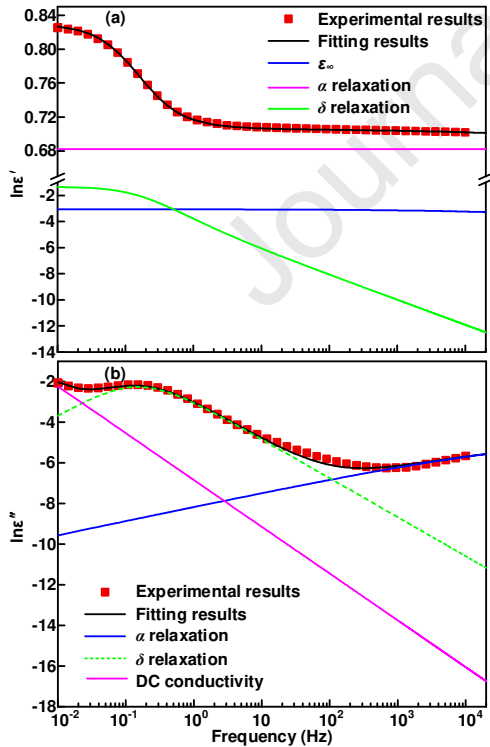


Figure 5. Experimental and fitting results for the real permittivity (a) and imaginary permittivity (b) of sPP0 at 70 °C. The curves and symbols denote the fitting and experimental results, respectively.

linear negative correlation between $\ln(\epsilon'')$ and frequency in the logarithm distribution is observed. Thus, this new process is attributed to a DC conductivity process at low frequency and mainly contributes to the dielectric loss [10, 12].

For more details on relaxation processes, the Havriliak–Negami (H-N) function equation is applied to fit the complex permittivity ϵ^* of α , δ and DC conductivity processes. Thus, some fitting parameters that are critical for the analysis of relaxation processes are available. The H-N equation can be written as

$$\epsilon^* = \epsilon_\infty + \frac{\Delta\epsilon_\alpha}{(1 + (i\omega\tau_\alpha)^{\beta_\alpha})^{\gamma_\alpha}} + \frac{\Delta\epsilon_\delta}{(1 + (i\omega\tau_\delta)^{\beta_\delta})^{\gamma_\delta}} + \frac{\sigma}{i\epsilon_0\omega} \quad (5)$$

where ϵ_∞ is the optical dielectric constant, which implies the polarization of atomic and electronic excitations when the frequency is very high. The relaxation time constant of ϵ_∞ is usually in the range of 10^{-16} - 10^{-15} s and 10^{-14} - 10^{-13} s [10]. ω is the electrical field oscillation frequency. For relaxation process α , $\Delta\epsilon_\alpha$ is the dielectric relaxation strength, τ_α is the relaxation time constant, β_α and γ_α are shape coefficients that describe the width and asymmetry of the relaxation spectra, respectively, while $\Delta\epsilon_\delta$, τ_δ , β_δ , and γ_δ are the similar parameters for dielectric relaxation δ . Finally, σ is related to the low-frequency DC conductivity, and ϵ_0 is the dielectric constant under vacuum. The H-N equation fits the five blend samples well at 50, 70 and 90 °C. The comparison between

experimental results and calculated ones of complex permittivity, taking sPP0 at 70 °C for example, is shown in Figure 5, indicating that the calculated curves fit the experimental ones well.

3.3 DC Breakdown Characteristics

The Weibull plots of the electrical breakdown test results are shown in Figure 6. E_b is the field strength at which there is a 63.2% probability for the sample to breakdown. The E_b is 360.2, 281.5, 325.4, 452.3 and 417.2 kV/mm for sPP0, sPP5, sPP15, sPP30 and sPP45, respectively. That sPP exhibits a better breakdown performance than iPP has been confirmed [5, 32]. The addition of sPP can first lower and then enhance the breakdown strength of PP samples, E_b reaches a maximum point at sPP30 and then decreases when the sPP content rises up to 45 wt%. The breakdown strength of virgin iPP is ~358 kV/mm [33], which is lower than that of sPP0, sPP30 and sPP45. The 5 wt% content of aPP has negligible effect on the reduction in the breakdown strength of PP blends, thus, the utilization of aPP for HVDC cable insulation is practical.

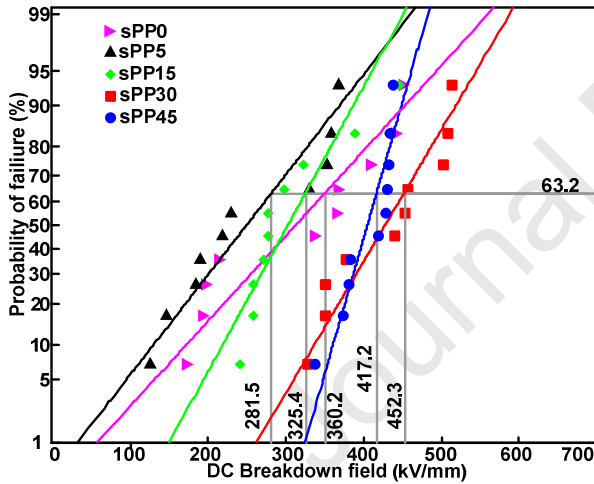


Figure 6. Electrical breakdown results of PP blends.

3.4 Trap level characteristics

The shallow and deep trap level and their density are presented in Figure 7. It is shown that the deep trap level and its density of sPP5 is lower than sPP0. Then the deep trap level, as well as its density, increases with the further increase of sPP content from 5 to 30 wt%. As the sPP content is up to 45 wt%, the deep trap level and its density decrease dramatically. The shallow trap level has the similar behaviours varying with the content of sPP. Nevertheless, the density of shallow traps generally do not change with the content of sPP. The trap level characteristics is closely related to the transport of charge carriers, and play an important role in the breakdown process, which will be discussed in the following section.

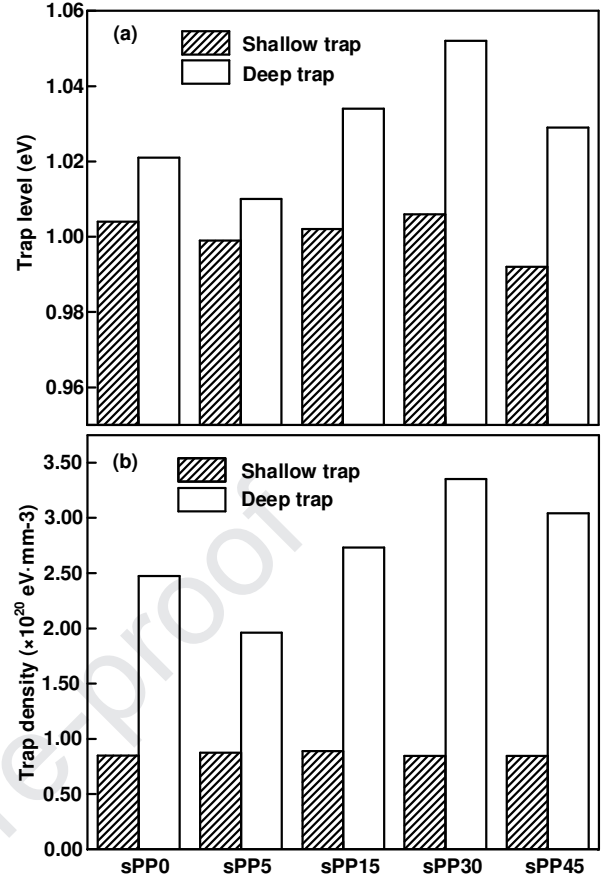


Figure 7. Trap centre densities (a) and energy levels (b) of deep traps and shallow traps of PP blends.

4. Discussion

4.1 Dipole Motion Relaxation

According to equation 5, the fitting results for τ_α , τ_δ , $\Delta\epsilon_\alpha$, $\Delta\epsilon_\delta$ and σ_{DC} are obtained. The values of τ_α are between 10^{-9} and 10^{-4} s, which are in the time range for the orientational polarization of molecules [10]. The τ_δ is in the range of 10^{-1} ~ 10^2 s, which belongs to the time range of hopping polarization [10]. Figure 8 shows the α relaxation strengths ($\Delta\epsilon_\alpha$) and δ relaxation strengths ($\Delta\epsilon_\delta$) of five samples at 50, 70 and 90 °C. $\Delta\epsilon_\delta$ increases with the temperature, since the thermal motion of charges is more intense, and the carrier mobility is enhanced in the PP bulk.

The α relaxation results from segmental motions in the amorphous phase of iPP, sPP and aPP [34], since the polarity of iPP, sPP and aPP molecules are similar. From the dipole point of view, α relaxation is a large-range dipolar motion relaxation contributed to by micro-Brownian segmental motions of polymer chains [35]. As the temperature rises, more dipoles gain mobility to orient with electrical field and contribute to the increase in permittivity. Therefore, $\Delta\epsilon_\alpha$ increases with the temperature. However, further increasing of the temperature will lead to even stronger Brownian

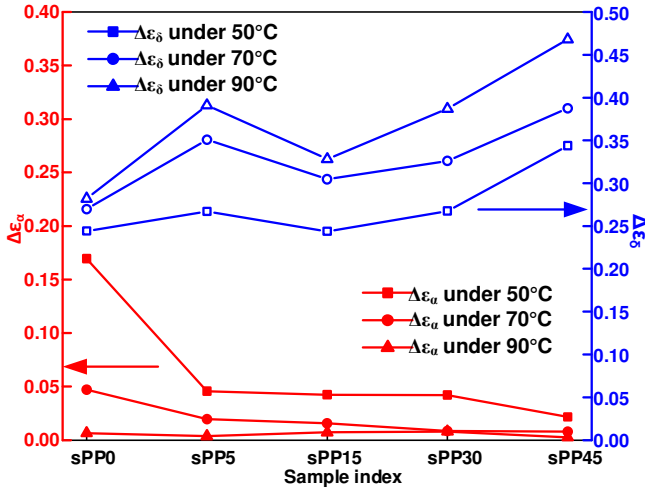


Figure 8. α Relaxation strength ($\Delta\epsilon_\alpha$) and δ relaxation strength ($\Delta\epsilon_\delta$) of PP blends at 50, 70 and 90 °C.

motion, causing the dipoles involved to undergo random thermal motion, thus preventing their complete orientation with the applied electrical field. Therefore, $\Delta\epsilon_\alpha$ decreases with temperature as illustrated in Figure 8. Moreover, in Figure 8, it can be found that $\Delta\epsilon_\alpha$ decreases as the amount of sPP increases. These phenomenon can be further investigated by the Kirkwood–Fröhlich equation [36]

$$\Delta\epsilon_\alpha = \frac{1}{3\epsilon_0} g \frac{\mu^2}{k_B T} \frac{N}{V} = \frac{1}{3\epsilon_0} g \frac{\mu^2}{k_B T} \frac{\rho_A N_A}{M} \quad (6)$$

where ϵ_0 is the dielectric constant under vacuum; g is the Kirkwood–Fröhlich correlation factor, which describes the molecular interaction between dipoles involved in the relaxation; μ is the mean dipole moment related to the process under consideration; N/V is the number density of dipoles contributing to the concerned dielectric relaxation; ρ_A is the mass density; M is the molecular weight of the material involved in the relaxation process; and N_A is Avogadro's constant. Since the α process is associated with dipoles in the amorphous region, ρ_A is the mass density of the amorphous PP. $\Delta\epsilon_\alpha$ decreases as the temperature rises based on equation 6, which is coincident with the above conclusions in this work.

Considering all other factors to be constant in equation 6, $\Delta\epsilon_\alpha$ can be simplified as

$$\Delta\epsilon_\alpha \propto g\mu^2 / T \quad (7)$$

Therefore, $\Delta\epsilon_\alpha$ is proportional to the mean dipole moment and intermolecular interaction at the same temperature. In addition, $\Delta\epsilon_\alpha$ can also be decided by the amount of dipole moments in the amorphous regions [35]. The samples with lower crystallinities contain more dipole moments in the amorphous regions and have higher $\Delta\epsilon_\alpha$. It is also assumed that the mean dipole moment (μ) of PP molecules does not vary in the PP blend systems, thus $\Delta\epsilon_\alpha$ is proportional to the product of the amount of dipole moments and correlation factor g [35, 36]. Referring to the crystallinity calculated in

3.1, there is no specific coincidence with $\Delta\epsilon_\alpha$ and the number of dipole moments. Thus, the amount of dipole moments can be concluded a minor factor, and the intermolecular interaction represented by g mainly decides $\Delta\epsilon_\alpha$. In this way, further investigations of g are made, as shown below. Previous work [37] showed that the value of the interaction factors for iPP and sPP in an iPP/sPP blend are essentially zero. These interaction factors are calculated from the Flory-Huggins theory, which is used to estimate intermolecular interaction. Another work [38] also demonstrates that the effect of the intermolecular interaction between iPP and sPP exists only when the amount of sPP is small, and the intermolecular interaction effect will generally become zero with the addition of sPP. Accordingly, with the increasing sPP content, the interaction factor (g) of PP blends reduces and contributes to the decrease in $\Delta\epsilon_\alpha$.

4.2 Carrier Hopping Behaviours

The dielectric relaxation process δ is ascribed to hopping polarization [12]. In a dielectric material, localized charges (ions and vacancies, or electrons and holes) can hop out from site A and be trapped in neighbouring site B, leaving a positive charge in site A, and creating a negative charge in site B, by this way a dipole is formed. [10]. A large number of deep traps concentrate in the interfaces between amorphous and crystalline phases of PP [16-18]. When the trapping of charge carriers is took place especially in the interfaces between iPP and sPP regions or amorphous and crystalline phases, this process can be named as interfacial polarization or Maxwell/Wagner/Sillars polarization[10, 51].

To calculate the activation energy of hopping polarization, τ_δ as a function of temperature is well fitted by the Arrhenius equation [39].

$$\tau_\delta(T) = \tau_0 \exp\left(-\frac{E_\tau}{kT}\right) \quad (8)$$

where τ_0 is the pre-exponential factor, E_τ is the activation energy, and k is the Boltzmann's constant. The Arrhenius plot of the temperature dependence of the relaxation time (τ_δ) is shown in Figure 9. The fitting parameters E_τ of sPP0, sPP5, sPP15, sPP30, sPP45 are 0.86, 0.83, 0.92, 1.05 and 0.54 eV, respectively.

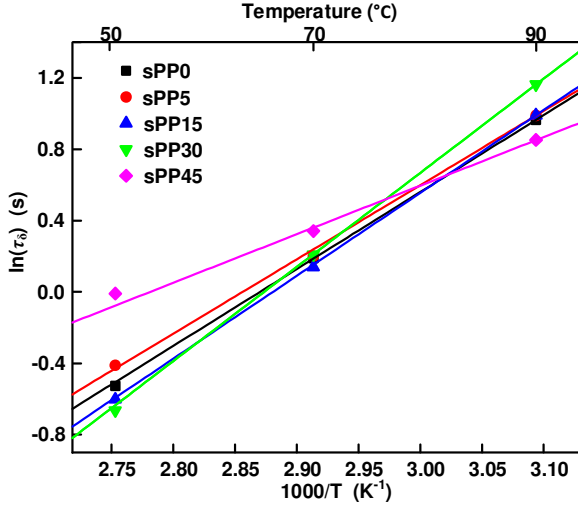


Figure 9. The relationship between the relaxation time constant ($\ln(\tau_\delta)$) of the δ process and the reciprocal of the temperature for PP blends.

The trap energy level indicates the potential barrier for the charge to hop from a localized state. Thus, there might be a correlation between hopping polarization and the trap characteristics. However, the effects of deep traps and shallow traps on hopping polarization can be significantly different. The deeper the trap, the longer the time that charges are detained in the trap. The difference in the trap filling time (τ_{trap}) might be several orders of magnitude, and this phenomenon is described by the equation [40]

$$\tau_{\text{trap}} = v_{\text{ATE}}^{-1} \exp(E_t / k_B T) \quad (9)$$

where v_{ATE} is the escape frequency of trapped charges, $v_{\text{ATE}} = k_B T / h$; E_t is the trap energy level, so for example, the trap filling time constant is approximately 10^{-13} s when the trap energy is 0.1 eV, while the time constant is approximately 500 s in 1.0 eV traps [41]. Therefore, it is barely possible for charges to escape from deep traps due to the extremely long relaxation time [42]. The carriers can hop out from shallow traps, but carriers in deep traps are unlikely to hop or tunnel between deep trap sites [13]. Accordingly, the correlation between shallow trap energy and activation energy of δ relaxation is investigated, as shown in Figure 10.

In Figure 10, the activation energy of the δ relaxation increases with the shallow trap energy. According to the hopping conductivity model, the effective carrier detrapping energy level (Φ_{eff}) in a dielectric is proportional to the trap level of the hopping states (Φ_t) [43].

$$\phi_{\text{eff}} = \phi_t - \frac{1}{2} e E_{\text{ext}} \lambda \quad (10)$$

where E_{ext} is the applied field, and λ is the mean free path of electrons. Namely, Φ_{eff} is the potential barrier for charges to surmount when they hop from localized site A. In equation 10, a deeper trap level (Φ_t) indicates a higher potential barrier (Φ_{eff}) for hopping process, thus, more activation energy is required for hopping polarization.

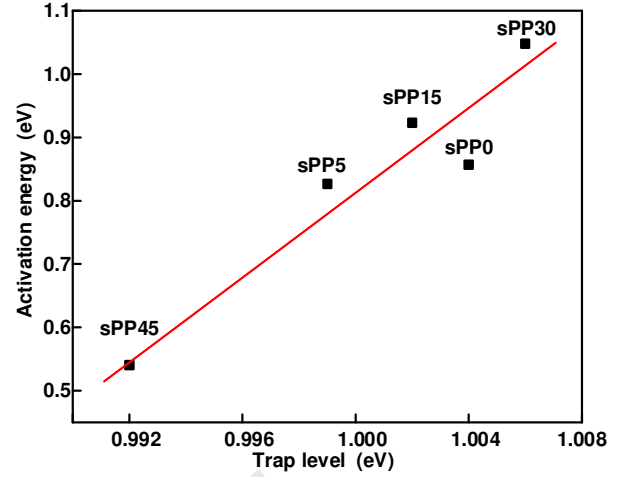


Figure 10. Activation energy of the δ relaxation as a function of the shallow trap level.

4.3 Correlation between Trap characteristics and Crystalline Behaviour

The correlation between trap density and crystallinity is shown in Figure 11. As the crystallinity rises, there is an uptrend exists for the deep trap density, while a downtrend is shown for the shallow trap density.

Carrier traps in polymer insulating materials, which are known as localized states present at a forbidden band, are considered to originate from chemical and/or physical defects within the polymer matrix [44, 45]. Deep traps mainly originate from chemical defects, while shallow traps mainly originate from physical defects. It has been widely confirmed that in semi-crystalline polymers, such as PP, shallow traps mainly exist in amorphous regions, while deep traps are concentrated at the boundaries between spherulites and the amorphous region [17, 33, 46].

The amorphous regions provide conformation disorders that offer diffusion channels through which carriers can easily penetrate [17]. As the crystallinity decreases, the amorphous content increases, and the number of physical defects caused by structural defects becomes enhanced, leading to a larger number of shallow traps [33].

The deep traps mainly concentrate along the boundaries between spherulites and amorphous regions. Previous research [30] has presented the morphology of PP blends by polarized optical micrographs and phase contrast micrographs, indicating that with the increase of sPP content, the spherulite size decreased, and the defects of spherulites increased. Thus, the larger interface regions in sPP30 and sPP45 specimens provide voids and cracks introduce more deep traps around spherulites that block the transport of charges [2, 16].

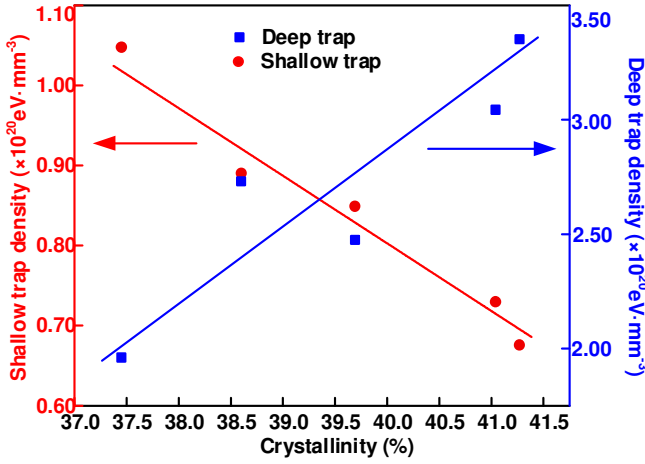


Figure 11. Relationship between the central trap density and the crystallinity of PP blends.

4.4 Effect of trap characteristics on breakdown behaviours

The trap density and level are considered effective at reflecting the breakdown performance of insulation materials from the microscopic point of view. Figure 12 shows the deep trap density and shallow trap density versus breakdown field. The plot indicates that the PP samples that contain more deep traps and fewer shallow traps exhibit higher breakdown strengths. Hence, the different effects of shallow traps and deep traps on the breakdown behaviour are investigated separately.

Assuming electrons are the majority carrier in the blend insulation. Electrons are accelerated in an applied field (E_{appl}), and an electron with free path λ can gain kinetic energy as $eE_{appl}\lambda$ [48-50]. A critical value for E_{appl} exists, under which the electron with the longest acceleration distance λ can cross over the trap barrier (E_t), when its kinetic energy $eE_{appl}\lambda$ is larger than that of the potential barrier E_t . The breakdown starts with the increasing number of such free charges and the more intense impact ionizations. The impact of molecules caused by such free charges can result in chain scissions and the formation of large molecule radicals, thus, low-density regions are formed in the amorphous area. As a result, the partial discharge starts in the bulk and further develops into the breakdown of polymer. On the contrary, a shorter length of free volume results in $eE_{appl}\lambda < E_t$, and the carriers will be captured by deep traps, thus the impact ionization and breakdown process are restrained.

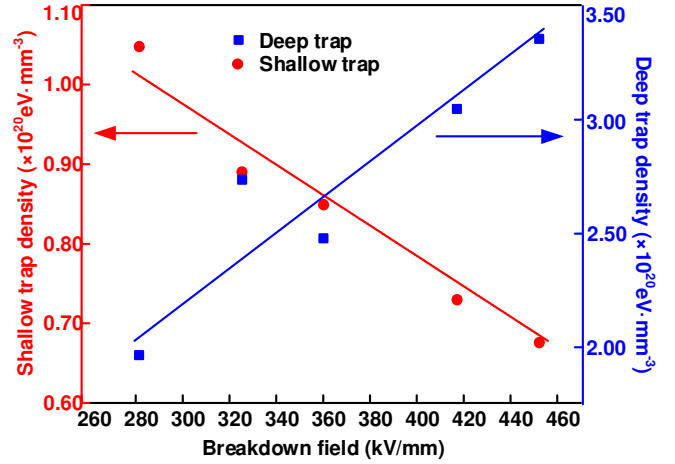


Figure 12. Deep trap density and shallow trap density as a function of the breakdown field.

This mechanism can be illustrated in Figure 13. The breakdown criterion is expressed by [49]

$$E_{critical} = \frac{E_t}{e\lambda_e} \quad (11)$$

Therefore, the critical breakdown field ($E_{critical}$) depends on the longest electron mean free path (λ_e) and the potential barrier (E_t), which refers to the deep trap level in this work [13, 50]. The deep traps have a significant effect on the charge transport [15]. The increased deep trap density (N_t) can increase the charge trapping probability and lower the mean free path, so that the charge trapping probability and mean free path are given as [15, 50]

$$P_{tr} = eN_t\mu_0 / \epsilon_0\epsilon_r \quad (12)$$

$$\lambda_{free} = \mu_0 E_{appl} \tau_{free} = 4\epsilon_0\epsilon_r E_{appl} / eN_t \quad (13)$$

where μ_0 is the carrier mobility, and τ_{free} is the mean free transport time of carriers, which is equal to the reciprocal of the charge trapping probability. In equations 11 and 13, the mean free path of charges increases with the decrease in deep trap density, thus, the charges can gain more kinetic energy in the field and cross over the deep traps, resulting in impact ionization and the reduction in breakdown strength. Similarly, the increased deep trap level (E_t) enhances the critical breakdown field (as shown in equation 11), since it leads to a higher potential barrier for the transport of free charges and the reduction in carrier mobility.

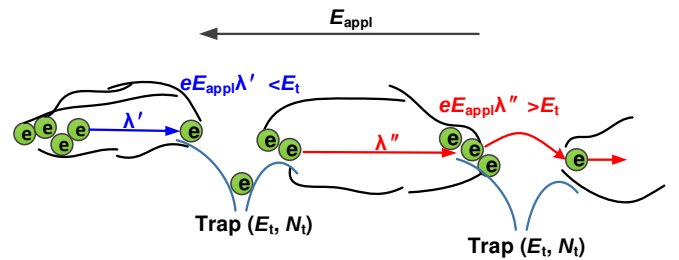


Figure 13. Schematic of the breakdown criterion.

5. Conclusion

In this work, iPP/sPP/aPP blends are prepared to investigate the effect of various sPP contents on the dielectric relaxation and crystallinity. Crystallinity is used to estimate the origin of traps in PP blends. The relationship between dielectric relaxation and the trap characteristics of PP blends is analysed, as well as their effect on breakdown properties.

The main results and conclusions can be summarized as follows:

1. α and δ relaxation processes in PP blends are proven to involve with segmental motion and carrier hopping, respectively. $\Delta\epsilon_a$ decreases with increasing content of sPP, since the intermolecular interaction factor (g) between PP molecules is weakened. The activation energy of carrier hopping polarization is proportional to the shallow trap level, since an increased shallow trap level can pose a higher potential barrier for carrier hopping.

2. A higher crystallinity results in a higher deep trap density and a lower shallow trap density in PP blends. Therefore, deep traps can be concluded to be induced by the weak points and voids at the boundaries of perfect PP spherulites, while shallow traps are mainly distributed in amorphous regions.

3. PP blends with higher deep trap densities and lower shallow trap densities show higher breakdown strengths. The increased deep trap density serves a function to reduce the mean free path of charges and restrain impact ionization, resulting in an enhancement in the breakdown field.

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Declaration of interests

☒ The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

☐ The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: